Software Test Plan (STP)

SMART COMPLAINT MANAGEMENT SYSTEM

Project: Smart Complaint Management System (SCMS)

Version: 1.0

Group No.: 8

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Team

Name	SRN	Role / Focus Area
Keertana M	PES2UG24CS638	QA Lead & Admin Complaint Management Testing
Raksha	PES2UG24CS652	Test Engineer — Complaint Tracking & User Status
Dhruva Ganesh Hegde	PES2UG24CS671	Test Engineer — User Account & Complaint Submission
ShivaSwaroop D S	PES2UG24CS679	Test Engineer — Complaint Resolution & Closure

Revision History

Version	Date	Author	Change Summary	Approval
1.0	16 September 2026	Group 8 (Keertana M, Raksha, Dhruva Ganesh Hegde, ShivaSwaroop D S)	Initial Software Test Plan for SCMS, based on SRS v1.0	—

Approvals (Summary)

Role	Name	Signature / Email	Date
Course Coordinator			

1. Introduction

Purpose: This document defines the test plan for the Smart Complaint Management System (SCMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers SCMS features such as user registration and authentication, complaint submission, complaint tracking, administrator complaint management, and resolution/closure. Integration with external third-party complaint systems and production-scale infrastructure testing are excluded.

References: SCMS SRS v1.0, Software Engineering course guidelines (UE24CS341A), PES University project submission standards.

Definitions: SCMS (Smart Complaint Management System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), FR (Functional Requirement), NFR (Non-Functional Requirement), UAT (User Acceptance Testing).

2. Test Items

- User Account Module (registration, login, logout, protected access)
- Complaint Submission Module (complaint form, validation, complaint ID, data storage)
- Complaint Tracking Module (complaint list, tracking, status and update display)
- Admin Complaint Management Module (admin login, complaint list/details, status updates)
- Resolution & Closure Module (resolution/response entry, closure, final user visibility)
- System-wide Controls (security, performance, reliability, usability, availability, maintainability)

3. Features to be Tested

Features mapped to SRS requirement IDs:

Req ID	Feature	Priority
SCMS-F-001	User registration with required details	High
SCMS-F-002	User authentication (login)	High
SCMS-F-003	User logout	Medium
SCMS-F-004	Complaint submission with mandatory-field validation	High
SCMS-F-005	Unique complaint ID generation	High
SCMS-F-006	Complaint detail storage	High
SCMS-F-007	Mandatory-field validation before submission	High
SCMS-F-008	View submitted complaints	High
SCMS-F-009	Track current complaint status	High

Req ID	Feature	Priority
SCMS-F-010	Display current complaint status	Medium
SCMS-F-011	Display updates/responses on a complaint	Medium
SCMS-F-012	Administrator login	High
SCMS-F-013	Admin views submitted complaints	High
SCMS-F-014	Admin views complaint details	High
SCMS-F-015	Admin updates complaint status	High
SCMS-F-016	Admin provides resolution/response	Medium
SCMS-F-017	Admin closes complaint	Medium
SCMS-F-018	User views final resolution and status	High
NF-001	Response time ≈ 2–3 seconds for common actions	High
NF-002	Availability during normal operating periods	Medium
NF-003	Usability — simple, clear interface	Medium
NF-004	Security — protection of sensitive information	High
NF-005	Reliability — no data loss on refresh	Medium
NF-006	Maintainability — modular structure	Low
SR-001	Protected access requires authentication	High
SR-002	Admin functions restricted to administrators	High
SR-003	Password hashing / no plaintext storage	High
SR-004	Cross-user complaint data isolation	High
SR-005	Administrator action recording	Medium

4. Features Not to be Tested

- Integration with external government, university, municipal, or third-party complaint systems
- Production-scale deployment and infrastructure testing beyond the project demonstration environment
- Use of real personal or sensitive complaint information (controlled dummy data will be used instead)
- Features not present in the approved SRS unless formally added to the project baseline

5. Test Approach / Strategy

Levels

- Unit / Module tests — verify individual functions and module behaviour
- Integration tests — verify interfaces and data flow between modules
- System tests — verify the complete end-to-end SCMS workflow
- Acceptance tests (UAT) — confirm that major project requirements are satisfied

Types

- Functional testing (core features, positive and negative scenarios)
- Regression testing (rerun relevant tests after fixes and changes)
- Performance testing (response time, concurrent-user load)
- Usability testing (UI clarity, navigation, validation messaging)
- Security testing (authentication, authorization, data isolation)

Entry Criteria: A testable build is available; the database, application services, and test accounts are ready; the relevant requirement and test case have been reviewed; test data is prepared and contains no real sensitive information.

Exit Criteria: All planned high-priority tests have been executed; no unresolved critical defect remains; important failed tests have been retested after fixes; requirement traceability and execution records are complete for the submitted build.

5.1 Security Validation

- Validate that protected routes and functions cannot be accessed without authentication
- Verify that admin-only functions are denied to normal user accounts
- Verify that passwords are stored in hashed form and never logged or displayed in plain text
- Verify that a user cannot view or modify another user's complaint data
- Verify that administrator status-update and resolution actions are recorded

6. Test Environment

Hardware: Laptop/desktop with a modern web browser (client side).

Software: SCMS web application v1.0, backend/API environment used by the team, and the project database.

Tools: Git/GitHub (version control), Jira (defect and task tracking), Postman or equivalent (API testing), and a suitable performance/load testing tool where required.

Test Data: Dummy user accounts, complaint records covering multiple statuses, and a valid administrator account, alongside a normal-user account used to test authorization boundaries.

7. Test Schedule

Milestone	Planned Date / Window
Requirement review and test-plan baseline	Week 1
User account and complaint submission testing	Week 2
Complaint tracking and user-status testing	Week 3
Admin complaint management testing	Week 4
Resolution, response, and closure testing	Week 5
Full integration and end-to-end system testing	Week 6
Regression, security, NFR, and acceptance testing (UAT)	Final week

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual)
- Test Data (dummy accounts, complaints, administrator credentials)
- Test Execution Records
- Defect Reports
- Test Summary / Acceptance Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Keertana M	Prepare the test plan, coordinate execution, and own admin login/complaint listing/status-update test cases.
Test Engineer (Complaint Tracking)	Raksha	Design and execute complaint list, tracking, and status/update display test cases.
Test Engineer (User Account)	Dhruva Ganesh Hegde	Design and execute registration, authentication/logout, and complaint submission test cases.
Test Engineer (Resolution & Closure)	ShivaSwaroop D S	Design and execute resolution, response, closure, and final-outcome test cases.
Course Coordinator	—	Review test results and approve project readiness for submission.

10. Risks and Mitigation

Risk	Mitigation
Delay in module completion due to overlapping academic workload	Prioritize high-priority modules first and track progress against the weekly schedule.
Limited or shared local test environment availability	Schedule testing slots among members and keep a backup local setup.
Integration testing depends on all four members' modules being ready	Maintain clear weekly checkpoints and use stubs/mock data for modules still in progress.
Incomplete or unrealistic test data	Prepare dummy accounts, complaints, and administrator credentials in advance during Week 1.

11. Assumptions & Dependencies

- Users provide the required information while registering and submitting complaints.
- Administrators have authorized access and are responsible for reviewing and processing complaints.
- Reliable storage and availability of complaint information is assumed throughout testing.
- Test data (dummy accounts, complaints, administrator credentials) will be prepared before execution begins.
- The application, database, and related services remain available during the scheduled testing weeks.

12. Suspension & Resumption Criteria

Suspend testing if:

- The application or database environment is unavailable for an extended period during a scheduled testing session.
- The build is unstable enough to block a large proportion of planned test cases.

Resume testing if:

- Blocking defects are resolved and a stable build is available.
- Environment access is restored.

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) ensures mapping of SRS requirements to test cases. Example:

- SCMS-F-001 (User registration) → TC-001
- SCMS-F-002 (User authentication) → TC-002
- SCMS-F-004 (Complaint submission) → TC-003
- SCMS-F-005 (Unique complaint ID) → TC-004

- SCMS-F-008–SCMS-F-011 (Complaint tracking) → TC-006
- SCMS-F-012–SCMS-F-015 (Admin complaint management) → TC-007
- SCMS-F-016–SCMS-F-018 (Resolution & closure) → TC-008, TC-009
- NF-001 (Response time) → TC-012
- SR-002 / SR-004 (Authorization / data isolation) → TC-010, TC-011

The complete requirement-to-test-case mapping is maintained in the Requirements Traceability Matrix section of the SCMS SRS.

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Weekly execution status (aligned with the weekly testing schedule)
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
Team Member — Admin Complaint Management	Keertana M	
Team Member — Complaint Tracking & User Status	Raksha	
Team Member — User Account & Complaint Submission	Dhruva Ganesh Hegde	
Team Member — Complaint Resolution & Closure	ShivaSwaroop D S	
Course Coordinator		
Faculty / Project Evaluator		

16. Requirements Traceability Matrix (RTM)

Status: N = Not executed, P = Passed, F = Failed, A = Automated.

Req ID	Requirement (short)	Module	Test Case(s)	Status
SCMS-F-001	Register user	User Account Module	TC-REG-01	N
SCMS-F-002	Authenticate user	User Account Module	TC-AUTH-01	N
SCMS-F-003	Logout	User Account Module	TC-AUTH-02	N
SCMS-F-004	Lodge complaint	Complaint Submission Module	TC-CMP-01	N
SCMS-F-005	Generate unique complaint ID	Complaint Submission Module	TC-CMP-02	N
SCMS-F-006	Store complaint details	Complaint Submission Module	TC-CMP-03	N
SCMS-F-007	Validate mandatory fields	Complaint Submission Module	TC-CMP-04	N
SCMS-F-008	View submitted complaints	Complaint Tracking Module	TC-TRK-01	N
SCMS-F-009	Track complaint status	Complaint Tracking Module	TC-TRK-02	N
SCMS-F-010	Display complaint status	Complaint Tracking Module	TC-TRK-03	N
SCMS-F-011	Display updates/responses	Complaint Tracking Module	TC-TRK-04	N
SCMS-F-012	Admin login	Admin Complaint Management Module	TC-ADM-01	N
SCMS-F-013	Admin views complaints	Admin Complaint Management Module	TC-ADM-02	N
SCMS-F-014	Admin views complaint details	Admin Complaint Management Module	TC-ADM-03	N
SCMS-F-015	Admin updates status	Admin Complaint Management Module	TC-ADM-04	N
SCMS-F-016	Admin provides resolution	Resolution & Closure Module	TC-RES-01	N
SCMS-F-017	Admin closes complaint	Resolution & Closure Module	TC-RES-02	N
SCMS-F-018	User views final resolution	Resolution & Closure Module	TC-RES-03	N

Req ID	Requirement (short)	Module	Test Case(s)	Status
NF-001	Response-time target	System-wide — Performance	TC-Perf-01	N
NF-002	Availability	System-wide — Reliability	TC-Avail-01	N
NF-003	Usability	System-wide — UI	TC-UX-01	N
NF-004	Security (data protection)	System-wide — Security	TC-Sec-01	N
NF-005	Reliability (data retrievable)	System-wide — Reliability	TC-Rel-01	N
NF-006	Maintainability	System-wide — Codebase	TC-Maint-01	N
SR-001	Protected access requires auth	Security Controls	TC-Sec-02	N
SR-002	Admin-only function restriction	Security Controls	TC-Sec-03	N
SR-003	Password hashing	Security Controls	TC-Sec-04	N
SR-004	Cross-user data isolation	Security Controls	TC-Sec-05	N
SR-005	Admin action recording	Security Controls	TC-Sec-06	N